TRITA-ITM-EX 2026:161
Stockholm, Sweden 2026
www.kth.se

Degree project in Engineering Design
First cycle, 15 credits

Smart Reusable Retail Security Tag

A tag that will revolutionize today's retail market

ESKIL BRATAAS MORIN AND NOA SAYELER



Kandidatexamensarbete TRITA-ITM-EX 2026:161

Smart Reusable Retail Security Tag

Eskil Brataas Morin

Noa Sayeler

KTH Industriell teknik och management

Godkänt	Examinator Stefan Ståhlgren	Handledare Stefan Ståhlgren
	Uppdragsgivare Friedrich Elsesser	Kontaktperson Stefan Ståhlgren

Sammanfattning

Idag använder klädbutiker separata system för stöldskydd, prismärkning och utcheckning. Detta leder till onödigt pappersavfall, långsammare kassabetalningar och osynkad lagerdata. Projektet initierades av Friedrich Elsesser, som identifierade problemet när han arbetade på klädbutiken Nudie Jeans. Utvecklingen av RFID- och e-pappersteknik har öppnat en ny värld full av smartare lösningar med stor potential att förbättra och ersätta dagens system. Projektet delades in i olika faser, med början i en analys, följt av idégenerering och konceptutveckling. Detta ledde tillsammans med en vidareutveckling, till en slutprodukt som uppfyller de behov som identifierats av målgruppen. Under projektets gång användes flera metoder som strukturerade arbetet, stödde planeringen och uppmuntrade till kreativitet. Slutresultatet presenteras som Smart Reusable Retail Security Tag, en produkt designad för att modernisera marknaden och konkurrera med dagens föråldrade lösningar.

KTH Industrial Engineering and Management

Degree project TRITA-ITM-EX 2026:161

Smart Reusable Retail Security Tag

Eskil Brataas Morin

Noa Sayeler

Approved	Examiner Stefan Ståhlgren	Supervisor Stefan Ståhlgren
	Commissioner Friedrich Elsesser	Contact person Stefan Ståhlgren

Abstract

Today, fashion retailers use separate systems for theft prevention, price labeling, and checkout scanning. This leads to unnecessary paper waste, slower checkouts, and disconnected inventory data. This project was initiated by Friedrich Elsesser, who identified the problem while working at the fashion retailer Nudie Jeans. The development of RFID and e-paper technology has opened a new world full of smarter solutions, and with a strong potential to improve and replace today's systems. The project was divided into different phases, starting with a detailed analysis followed by idea generation and concept development. This, combined with further development, led to a final product that meets the needs identified by the target group. During the project, several methods were used to structure the work, support planning, and encourage creativity. The final result is presented as the Smart Reusable Retail Security Tag, a product designed to modernize the market and compete with today's outdated solutions.

FOREWORD

The group would like to thank the following people

Stefan Ståhlgren for guiding us and giving us invaluable advice along the way.

Friedrich Elsesser for giving us this great opportunity to work with him and for sharing his competence with us throughout the entire process.

Vusion Group, for kindly participating in an interview and providing insights that were very valuable for the development of the project.

The prototype workshop at KTH for the usage of machines and materials.

Eskil Brataas Morin and Noa Sayeler

Stockholm, 22 May 2026

NOMENCLATURE

Abbreviations

RFID	Radio Frequency Identification
NFC	Near Field Communication
OWN	Optical Wireless Network
ESL	Electronic Shelf Label
EAS	Emergency Alert System
LCA	Life Cycle Assessment

TABLE OF CONTENTS

SAMMANFATTNING ii

ABSTRACT iv

FOREWORD vi

NOMENCLATURE viii

TABLE OF CONTENTS x

1 INTRODUCTION 1

1.1 Background 1

1.2 Aim and purpose 1

1.3 Constraints 1

1.4 Method 1

2 ANALYSIS 3

2.1 Market research 3

2.1.1 Online research 3

2.1.2 Store visits 3

2.1.3 Interviews 4

2.1.4 Surveys 4

2.2 Target audience analysis 5

2.2.1 Target audience 5

2.2.2 User needs 5

2.2.3 Journey mapping 5

2.3 Moodboard 6

2.4 Future and sustainability analysis 7

2.5 Requirements specification 8

3 IDEA GENERATION 9

3.1 Brainstorming 9

3.2 Braindrawing 10

3.3 Scamper 10

3.4 Physical models 11

4 CONCEPT DEVELOPMENT 13

4.1 Concept generation 13

4.1.1 Concept 1 (The Base) 13

4.1.2 Concept 2 (The Minimalist) 13
4.1.3 Concept 3 (The Terminator) 14
4.2 Concept selection 15
4.3 Refinement of selected concept 15
4.3.1 Front 15
4.3.2 Back 16
4.3.3 Other 16
5 MATERIAL RESEARCH 17
5.1 Materials 17
5.2 Material selection 18
6 COST AND WEIGHT ANALYSIS 19
7 RESULTS 21
7.1 Final product 21
7.2 Evaluation against requirements specification 23
7.3 Manufacturing 23
8 DISCUSSION 25
9 CONCLUSION 27
10 SOURCES 29
11 APPENDIX 30
11.1 Appendix A: Store visits 30
11.2 Appendix B: Requirement specifications 31
11.3 Appendix C: Revisits to stores 32
11.4 Appendix D: Visibility test 33
11.5 Appendix E: Validation of the screen angle 34
11.6 Appendix F: Testing the back component 35
11.7 Appendix G: Analysis of internal layout 36
11.8 Appendix H: Cost and weight calculation 37
11.9 Appendix I: Requirements specification validation 38
11.10 Appendix J: Manufacturing 40

1 INTRODUCTION

This chapter introduces the project, including its background, purpose, and constraints. It outlines the motivation behind the project and presents the problem statement that guided the development process.

1.1 Background

The project was given by Friedrich Elsesser. While working at Nudie Jeans in Stockholm, he noticed a problem with the current system employed at the store. Most stores use three different systems for theft protection, price and product information, and checkout scanning, leading to paper waste, slower checkouts, and disconnected inventory data. New rising technologies such as RFID and e-paper technology can solve these problems, leading to streamlined retail operations.

1.2 Aim and purpose

The purpose of this project is to develop a product that solves the problems identified in the analysis. The following problem statement guides the project:

How do we create a smooth, modern solution that integrates theft protection, price management, and cash register systems to streamline store operations and minimize errors, while maintaining a premium feel for customers?

1.3 Constraints

To ensure the project is completed within the established timeframe, certain limitations have been imposed. The electrical routing and circuits have been excluded and are not a part of the project. However, the group has found suitable components and made sure they fit, but the internal placement has not been decided. As these technical problems lie beyond the group's area of expertise, it was decided to be excluded from the project.

1.4 Method

The project consisted of two parts: an analysis phase and a concept development phase.

In the analysis phase, a Gantt chart was created along with a plan for how the group would structure the work going forward. Research, observations, and interviews, together with the requirements specification, formed the base for decisions and evaluations throughout the project.

In the concept development phase, the focus was on generating ideas to create a wide range of solutions. This was done using methods such as braindrawing, scamper, and developing models, along with multiple iterations to ensure that the final solution was strongly based on the findings from the analysis phase.

2 ANALYSIS

During the analysis phase, comprehensive research was conducted to evaluate the current market. The goal was to identify existing needs, successful market solutions, and potential areas for development.

2.1 Market research

To develop a product that meets customer needs, market research was carried out through methods such as online research, store visits, interviews, and surveys.

2.1.1 Online research

The online research was focused on exploring existing anti-theft solutions currently used in the industry, as well as looking at emerging technologies such as RFID and E-paper.

Regarding emerging technologies, three solutions were identified: RFID, NFC, and OWN.

RFID is currently the most widely used technology in retail due to its versatile applications, according to Atlas RFID (n.d). The technology works by sending out radio waves that communicate with an RFID tag. The tag sends back a unique signal that helps the system identify the item. According to Huayuan (n.d.), Uniqlo has successfully used RFID technology in its stores. At checkout, RFID readers in the checkout bin can detect all clothing tags at the same time, instead of scanning each barcode one by one. This has streamlined the checkout process, making it faster.

The second technology studied was NFC, which allows secure data transfer over short distances. For it to work, the transmitter and receiver need to be close to each other. A common example of NFC in everyday life is Apple Pay (Wikipedia, 2025).

Finally, OWN was researched. This technology uses infrared light signals that move too fast for the human eye to see. The signals bounce between the transmitter and the receiver to send information. One limitation is that the signals cannot pass through solid objects, which means the transmitters need to be placed in the ceiling (Pricer, n.d.).

2.1.2 Store visits

Several store visits were conducted to gain insights into current solutions. The following stores were visited:

Nudie Jeans, Uniqlo, Stadium, Clas Ohlson, BasketShop.se, Patagonia, Peak Performance, LIDL, and Arket.

During these visits, different parts of current retail systems were studied, including how product information is handled, how labels are used, and how checkout systems work. Photos of important features that influenced the project can be found in Appendix A.

Some visits gave more useful information than others, but all of them helped the project move forward. They all contributed to a deeper understanding of the project, and at the same time showed a proof of concept. The visit to Nudie Jeans was particularly important, as it was there that the idea came to life. Discussions with the staff provided valuable insights, including a better

understanding of how the supply chain operates, how they work with labeling, and the challenges they experience with their current solutions.

2.1.3 Interviews

Two interviews were carried out to better understand the users and the technical side of the project. The first interview was with an employee at Stadium, and the second with an engineer at VusionGroup, a leading company working with ESL.

The interview at Stadium started after learning that the company won the “Logistics Solution of the Year” award in 2022. Their system uses RFID on all clothing labels, which increased stock accuracy from 65% to 99.7% according to Dagens Logistik (n.d). This improvement came from combining RFID technology with better work routines. Today, RFID labels are added in their central warehouse in Norrköping before products are sent out to stores. When the goods arrive, staff can scan full delivery boxes in a few seconds using RFID scanners to check that everything is correct. By using a “move” function in their system, stock is updated when items are moved from the warehouse to the store. The same method is used when items are moved from storage to the shop floor, helping maintain full control and reducing the number of missing items, thereby increasing stock accuracy.

The second interview was with Jauck Philipp from Vusion Group. The focus was on ESLs, especially the cost and how the technology works. A normal ESL costs about 4–6 EUR, and the price goes up mainly when the screen is larger or when it can show more colors. Jauck also explained that if the label is updated twice a day, the battery can last more than 10 years. He also mentioned that most systems use either coin-cell batteries or small battery packs.

He also explained how updates work. Vusion Group uses a 2.4 GHz network controlled by a central server, which sends data to the labels using radio signals. This makes updates fast and efficient. The system can also track ESLs location using Bluetooth, which may help reduce theft. Finally, he said that ESLs are built to be strong and can handle scratches, temperature changes, and magnetic fields.

2.1.4 Surveys

Surveys were sent out to better understand the target audience. By using various groups on Facebook, a total of 20 unique responses were collected. Some selected responses from the survey can be found in Figure 1.

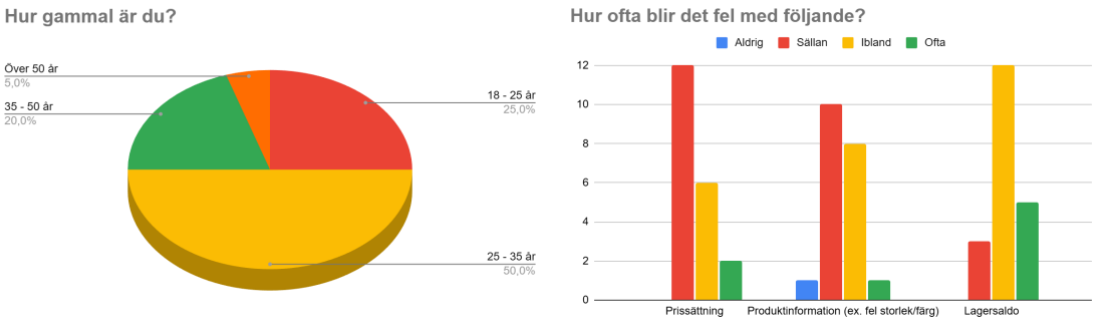


Figure 1. Selected responses from the survey

The surveys provided data which gave several key insights. Statements such as 'Inventory management is extremely time-consuming,' 'During large deliveries, we must manually inspect every item to verify the shipment; this can involve hundreds of garments,' and 'Prices are updated frequently' served as a proof of concept. These insights confirmed the market demand for a smart and efficient integrated solution.

2.2 Target audience analysis

A target audience analysis was done to understand who the users are and what they need. This helped in two ways: first, it guided the idea development, and second, it was used to check that the final ideas match what the users actually need.

2.2.1 Target audience

The target audience was split into two main groups to make sure all users are included. Store employees use the system every day, so the design needs to fit their daily work. Retail customers are also affected because the new solution replaces existing systems. Therefore, the two target groups are store employees and retail customers.

Store employees at retailers like Nudie Jeans are mostly between 25 and 35 years old, with more women than men. Working in a mid-premium store means they need to focus strongly on good customer service and have good knowledge about the products.

The second group is the customers. They are a wider age group, usually between 25 and 45 years old. Because the brand is premium, prices are higher, but customers accept this in return for better quality and a better shopping experience. This also means they usually have a middle to high income level.

2.2.2 User needs

Clear user needs have been identified to make the idea phase easier later in the process. These needs are categorized into three distinct groups to help distinguish between essential requirements and desirable features. The following needs have been extracted from all gathered data, and can be found in Table 1.

Table 1. User needs

Base	**Conscious**	**Unconscious**
Price information	Simplified inventory	Premium aesthetics
Product information	Different systems integrated	Durability
Anti-theft	Stock accuracy	Ergonomic
Daily updates	Easy to use	Feels modern
Lightweight	Easy to reuse	Tactile feel
Visible	Battery status	
Fastening mechanism		

2.2.3 Journey mapping

As described in the target audience section, this project has two main user groups: store employees and retail customers. Since they use the solution in different ways, separate user

journeys were created to show how each group interacts with it and where the key points of contact are.

2.2.3.1 The customer journey

The customer journey is divided into three main stages: arriving at the store, browsing and trying on clothes, and making the final purchase. When the customer enters the store, they form an early impression based on the store design, staff, and how products are shown. Together, these shape the brand experience.

Next, the customer looks around the store to find items they like. At this stage, product displays, available information, and prices play a big role in the experience. In many stores, unclear or wrong product information can cause frustration.

After choosing and trying on clothes, the journey ends with the purchase. However, the checkout process is often a friction point where queues and confusion can occur, especially if the price on the tag does not match the system price, leading to lower customer satisfaction.

2.2.3.2 The employee journey

The store employee's work includes inventory management, price updates, stock deliveries, price/product labeling, security tagging, customer service, checkout operations, and returns. The workday often begins with inventory counting, conducted daily, weekly, or monthly, which is not only extremely time consuming but also highly vulnerable to human error. Price updates are currently done manually, which becomes particularly challenging during major sales events such as Black Friday, due to the high volume of items that must be updated simultaneously. This makes the process extremely labor intensive. Following this, incoming deliveries must be inspected item by item to verify accuracy, which can be very time consuming when shipments consist of several hundred products. Once verified, each item must be unpacked, labeled with price and product information, fitted with a security tag, and finally placed on the sales floor.

The final stages of the employee's journey focus on floor operations. This begins with customer interaction, where common questions involve size availability and pricing. A frequent issue arises when items listed as 'in stock' online cannot be located in the store, leading to a poor customer experience. Furthermore, employees face high pressure and stress at the checkout counter due to peak hours and customer frustration when the price shown in store does not match the price registered in the system during payment. The journey concludes with the returns, ensuring that returned garments are in the correct condition and that the transaction is handled securely.

2.3 Moodboard

Moodboards are collections of images used to show a specific feeling, style, or aesthetic. These are used to facilitate the ideation phase and provide inspiration for the sketching process. In this project, two different moodboards were developed: one focused on emotion and feel, and the other on functionality.

As described in the Target Audience section, the solution is made for the mid-premium retail segment. Because of that, it must convey a sense of modernity while being perceived as premium, reliable, and robust. A moodboard for the overall feeling of the product can be found in Figure 2.



Figure 2. Moodboard for overall feeling

To fulfill the requirements previously defined, a moodboard focused on functionality was developed. This included everything from ESLs to diverse security tags. A moodboard for the functionality of the product can be found in Figure 3.



Figure 3. Moodboard for functionality

2.4 Future and sustainability analysis

A future and sustainability analysis aims to identify trends and factors that may influence the product.

The future analysis indicated that the primary focus lies in improving efficiency in processes. Companies continuously seek to minimize costs, and by constantly challenging existing solutions, they are able to achieve this. Modernization is also a key focus area, and implementing new technologies is an effective way to accomplish it. The growing market for RFID technology indicates how companies today are striving to modernize while simultaneously improving efficiency in their processes. This is clearly reflected in the annual increase in RFID adoption, which is estimated to be between 10–20% (Fortune Business Insights, 2026).

Regarding the sustainability analysis, the main focus is on improving the company's image toward greater sustainability. This is important both because it allows companies to present themselves as more sustainable and because sustainability has become an increasingly important issue for customers. According to Lindholm, A. (2026) 64% of today's consumers in the products and services sector are willing to pay more for a more sustainable product or service. This increases the pressure on companies to prioritize sustainability.

2.5 Requirements specification

To move on to the next phase, the analyses and identified needs were translated into clear requirements. The requirements specification is based on all work carried out before idea generation and is presented in Appendix B.

The appendix includes a column containing the letters "K" and "Ö", where "K" stands for requirements and indicates criteria that must be fulfilled, while "Ö" stands for desired features, meaning they are not mandatory but considered beneficial if achieved.

3 IDEA GENERATION

The idea generation process is divided into four phases, each building on the previous phase. This work is based on the analysis section and acts as a bridge between research and design.

3.1 Brainstorming

Brainstorming is the first of the four phases and focuses on generating a variety of ideas. At this stage, the focus is on generating many ideas rather than evaluating their quality, based on the principle that no idea is a bad idea. The brainstorming phase was conducted using a mind map. The mindmap is illustrated in Figure 4.

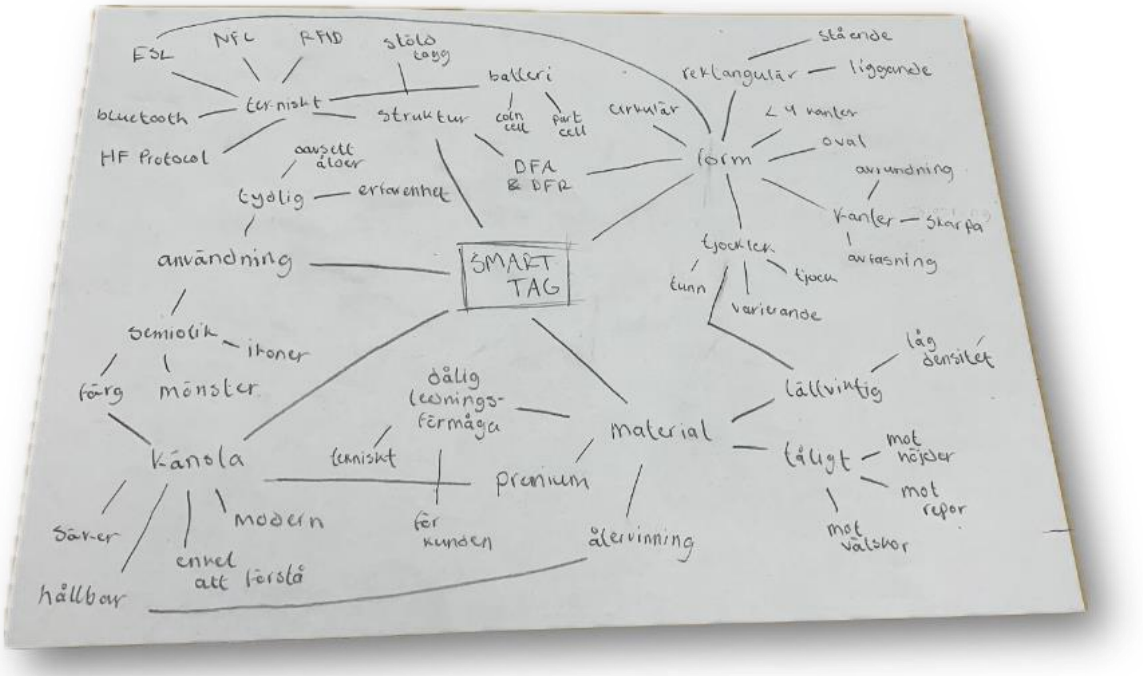


Figure 4. Mindmap

3.2 Braindrawing

Braindrawing was then used to better visualize the generated ideas. The focus of this phase was to translate ideas into simple sketches. This process was based on the mind map from the previous section and all ideas were documented, even if they were not technically possible or realistic. The results from applying Braindrawing are presented in Figure 5.

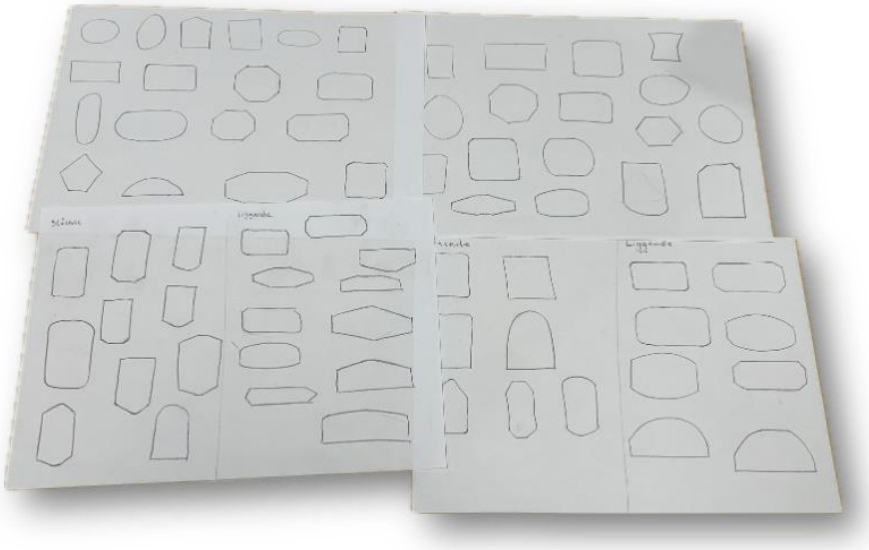


Figure 5. Braindrawing

3.3 Scamper

Scamper was the next phase in the ideation process, serving to further develop the previous phase. The most promising ideas from earlier stages were reviewed and further developed through applying the scamper technique. The results from applying Scamper are presented in Figure 6.

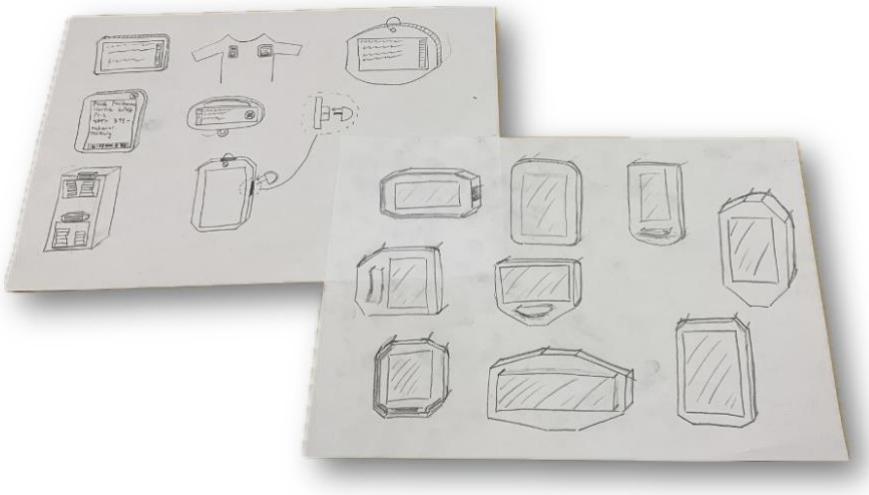


Figure 6. Scamper

3.4 Physical models

Since sketches alone can often make it difficult to visualize a product, several physical models were developed. Based on the sketches from the previous section, these models were created to help understand the actual dimensions. The physical models are shown in Figure 7.

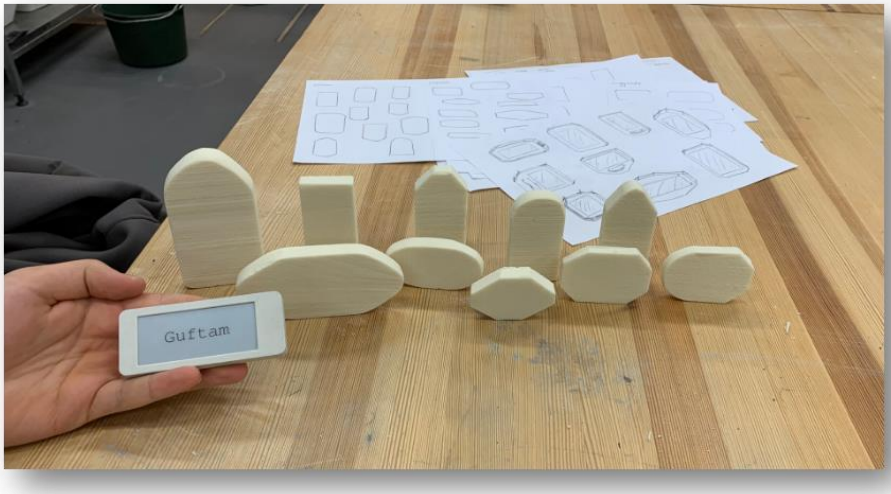


Figure 7. Physical models

4 CONCEPT DEVELOPMENT

In the following sections, the concepts developed are described.

4.1 Concept generation

Following an extensive ideation phase, three different concepts were developed. During this stage, the analysis and moodboards were used together with the requirements specification. The group decided that there would be two shared features across all concepts: the use of a 2.4 GHz network for updating the ESL data, and the integration of RFID technology to enable a more modern and efficient inventory management process.

4.1.1 Concept 1 (The Base)

The Base is the first of the three concepts and represents the original core idea. This concept fulfills all fundamental requirements, featuring integrated systems, a secure attachment mechanism, and a lightweight design. The checkout process in this case uses a RFID reader, similar to the systems currently implemented by retailers like UNIQLO and ZARA. The Base is shown in Figure 8.

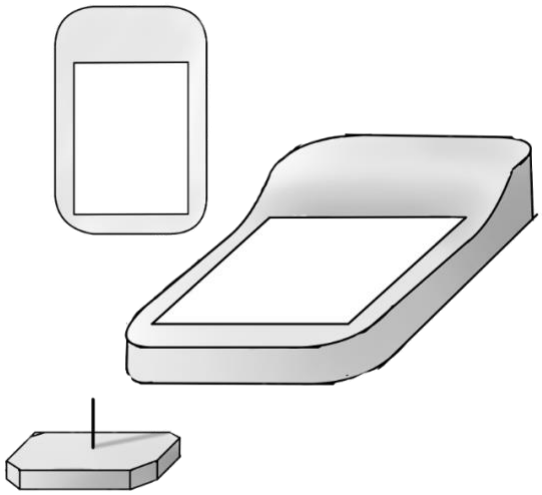


Figure 8. The Base

4.1.2 Concept 2 (The Minimalist)

The Minimalist is a variation of the previous concept, designed to be smaller and more compact. This concept aims to be cost efficient and only fulfil the most basic needs. The Minimalist is based on an observation made during the analysis phase. In retail stores, identical garment models are often grouped together. In those cases, the only unique attributes for each individual item are usually its size or color. Displaying repetitive information, such as the model name or brand on every single tag is unnecessary and can lead to higher production costs. The Minimalist

features a display that only shows the essential information for each garment, along with a barcode for checkout. The Minimalist is shown in Figure 9.

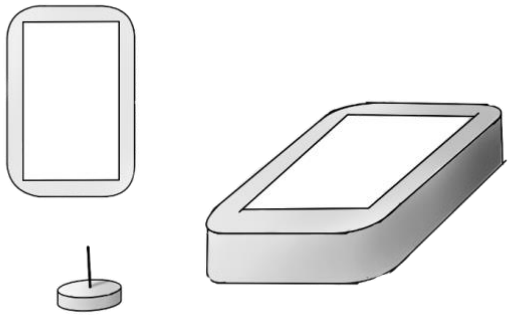


Figure 9. The Minimalist

A complement to this concept would be a larger product information display placed near the garments.

4.1.3 Concept 3 (The Terminator)

The Terminator, the third and final concept, is a maximized version of the original core idea. This concept focuses on latent needs to create a 'wow factor.'

During store observations, it was noted that security tags often get a natural tilt when attached to garments. If a screen is placed flat on such a tag, it becomes harder to read. To solve this, the screen is angled to improve readability and ergonomics. A volcano-shaped part is also added to create space for the security pin, similar to tags in the current market.

To make The Terminator feel modern and advanced, NFC technology is used at checkout and when removing the tag. An NFC chip is integrated in the tag and a reader in the detacher, making it possible to transfer data while removing the tag. The Terminator is shown in Figure 10.

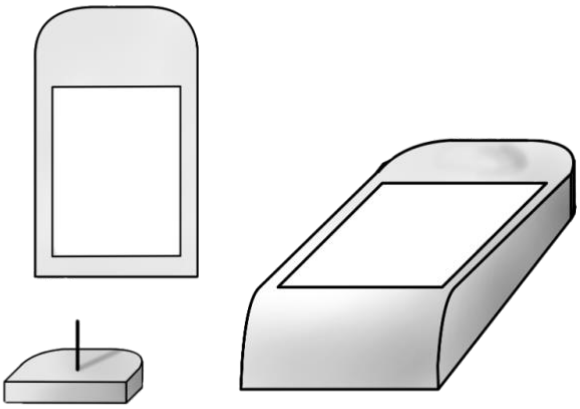


Figure 10. The Terminator

4.2 Concept selection

After performing a Criteria Weighting Matrix (see Figure 11), it was decided to proceed with the third concept: The Terminator.

Criterion	Weighting	The Base		The Minimalist		The Terminator	
		Rating	Total	Rating	Total	Rating	Total
Low weight	4	2	8	5	20	3	12
Clear reading	4	4	16	2	8	5	20
More efficient checkout	5	4	20	1	5	5	25
Premium feel	3	3	9	2	6	4	12
Manufacturing	2	3	6	5	10	2	4
Total:		59		49		73	

Figure 11. Weighted criteria matrix

4.3 Refinement of selected concept

The selected concept was further refined to optimize its design and functionality. This development phase was divided into two distinct focus areas: the front and the back.

4.3.1 Front

After analyzing the geometry, it was decided to round off the entire underside to create a more modern look. This also allows it to minimize material usage, helping reduce the overall weight.

The screen size was maximized based on two key factors: store revisits and visibility tests. During revisits to the stores, as presented in Appendix C, the size of labels used in retail environments was examined to gain an understanding of their dimensions and the type of information most commonly displayed. In addition, discussions with store employees provided further insights regarding size limitations. The minimum size was identified as 30 x 60 mm, while the maximum size was 70 x 100 mm. This helped provide an indication for the size of the product.

The second test was a visibility test. The visibility test, presented in Appendix D, was based on printing boxes containing the same information in different sizes on A4 paper. These were then viewed from a distance to assess whether the information could be clearly read or not. The test results led to increasing the screen size so that all customers can easily read the information.

To fulfill cognitive user needs, semiotics were applied by adding a grooved texture to the sides. This makes the product easier to understand and use, clearly showing how to handle the tag and giving a better grip during use.

After the final concept was selected, it was found that the initial design did not have enough internal space for the security pin. Therefore the bulge on the front was made more extreme to make space for the pin's length.

The angled design was a bold and innovative solution, and to validate the need for it, it was tested in the workshop. In Appendix E, tests were conducted by reading the display from different angles. The results showed that the concept is unique and solves an identified user need.

The technologies were also further investigated to ensure that they would function and solve a current problem.

4.3.2 Back

The focus during the development of the back was to simplify the attachment and detachment process. This was achieved through iterative testing of various sizes and textures to improve tactile feeling, grip, and how intuitive it is to use. After evaluating different shapes, a rounded profile was selected as the optimal solution for improved handling. The rounded design is more comfortable against the skin, especially when trying on tight clothes like jeans, where sharp edges could cause irritation. This was confirmed through tests with the product attached to garments worn in a real retail setting. These tests can be found in Appendix F.

Similar to the front part, the semiotics of the back were examined. Grooves were added to maximize grip and the understanding of where to hold the product.

4.3.3 Other

To continue developing this solution, it was essential to ensure that all components could fit within the product. The group investigated which components were required, partly for the cost and weight analysis described in a later section, but also to verify that all components could physically fit inside the product. In Appendix G, colored blocks were used to show the approximate sizes of the components, and the result showed that the concept would work.

5 MATERIAL RESEARCH

A material research study was conducted to identify the materials best suited for the solution.

5.1 Materials

Three different materials were evaluated: cork composites, recycled ABS plastic, and virgin plastic.

Cork composites provide natural shock absorption, helping protect the display and internal components. The material is lightweight, reducing strain on garments, while also offering a premium fee. Its low thermal conductivity improves tactile comfort, and its sustainable characteristics can improve a brand's environmental profile.

The material may have limited long-term durability, as it can wear down or crumble over time. Its porous structure can also absorb sweat, oils, and dirt, creating hygiene problems. In addition, cork composites can be difficult to manufacture with high precision, challenging to recycle due to mixed materials, and more expensive than traditional plastic alternatives.

Recycled ABS plastic is durable and resistant to cracking, even in thin or sensitive sections. The material is lightweight, reducing strain on garments, and can handle both high and low temperatures. It also works well for injection molding, allowing for detailed and design adapted shapes while contributing to a long product lifespan.

Recycled ABS plastic still has environmental limitations compared to fully sustainable materials. Its smooth surface can easily become scratched or visibly worn over time, negatively affecting its appearance. In addition, the material has a relatively hard, less premium tactile feel, and production costs may still be higher than simpler plastic alternatives.

Virgin plastic is cost effective for mass production and generally stronger and more durable than recycled plastic. The material is lightweight, reducing strain on garments, and is easier to manufacture with fewer risks of defects.

The material has a significant negative environmental impact and may result in additional EU fees related to non-recycled plastics. It can also scratch relatively easily and may create brand image risks due to sustainability concerns. Compared to cork composites, virgin plastic has a colder and less premium feel.

5.2 Material selection

After analyzing and considering all three materials, it was chosen to continue with recycled ABS plastic.

The material offers high durability and a long lifespan with a low risk of cracking even in thin sections, while its low weight minimizes the strain on the garment. In addition to being temperature resistant, the material is easy to manufacture through injection molding, enabling the detailed shapes required for the product. By prioritizing recycled plastic over virgin plastic, the negative environmental impact is reduced, strengthening the brand's environmental profile.

6 COST AND WEIGHT ANALYSIS

A cost and weight analysis are presented below to give an indication of the product's cost and weight.

A weight and cost calculation was performed to estimate the final product's weight and price. The weight calculation was based on the 1–3–9 model, where the price of all components represented the "1" part, this was then multiplied by 3 to get the production cost. The production cost of the final product amounted to 313.5 kr. The weight was calculated by adding the weights of all components, resulting in a total of 50.1 grams. These calculations should be seen more as indications rather than exact values. The cost and weight analysis can be found in Appendix H.

7 RESULTS

The final result of the project is presented below. The project resulted in a solution shaped by the requirements set by the commissioner as well as the needs identified during the market research.

7.1 Final product

The final product consists of two main parts: the front and the back. The front serves as the main component and displays price and product information via an e-paper display. The screen, as previously described, is wirelessly updated via a 2.4 GHz network, enabling fast and efficient updates. RFID technology is integrated into this component to improve inventory management. In addition, an NFC chip is integrated into the bulge of the tag. At checkout, when removing the tag from the garment, it allows you to transfer data while removing the tag, completely removing the need for a barcode displayed on the screen.

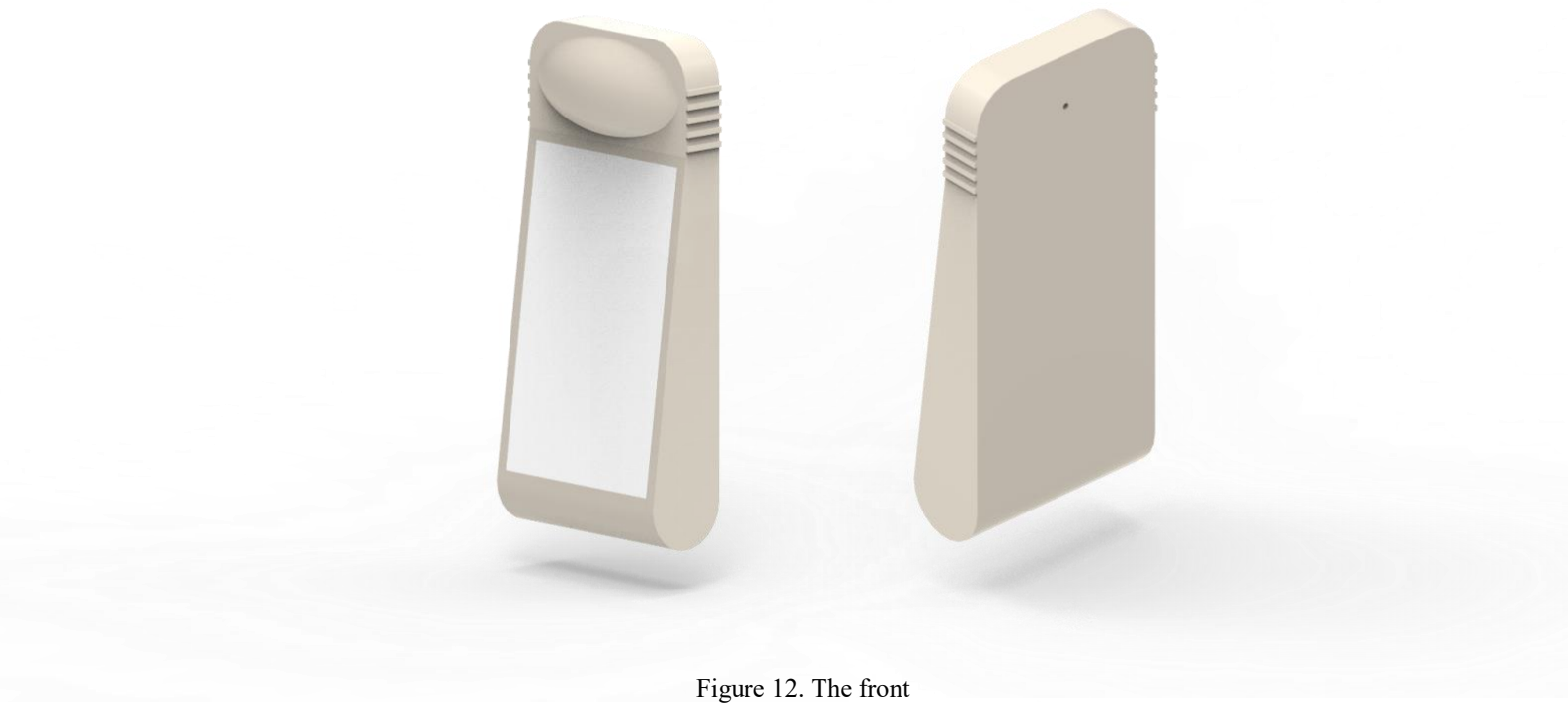


Figure 12. The front

Figure 12 illustrates the exterior, designed both to create a more polished final product and improve its usability. The design is ergonomically optimized with an angled screen, which compensates for the natural tilt that occurs when security tags hang on garments. Additionally, the grooves located next to the bulge are intended to improve the user's understanding of where to hold the product while also increasing grip. The bulge is designed to create space for the pin coming from the back part, which is used to attach the front and back sections.

The back is the part that is attached using a magnetic locking mechanism to secure the entire product onto the garment. To attach the product, a pin is used, similar to those found in today's EAS anti-theft tags.

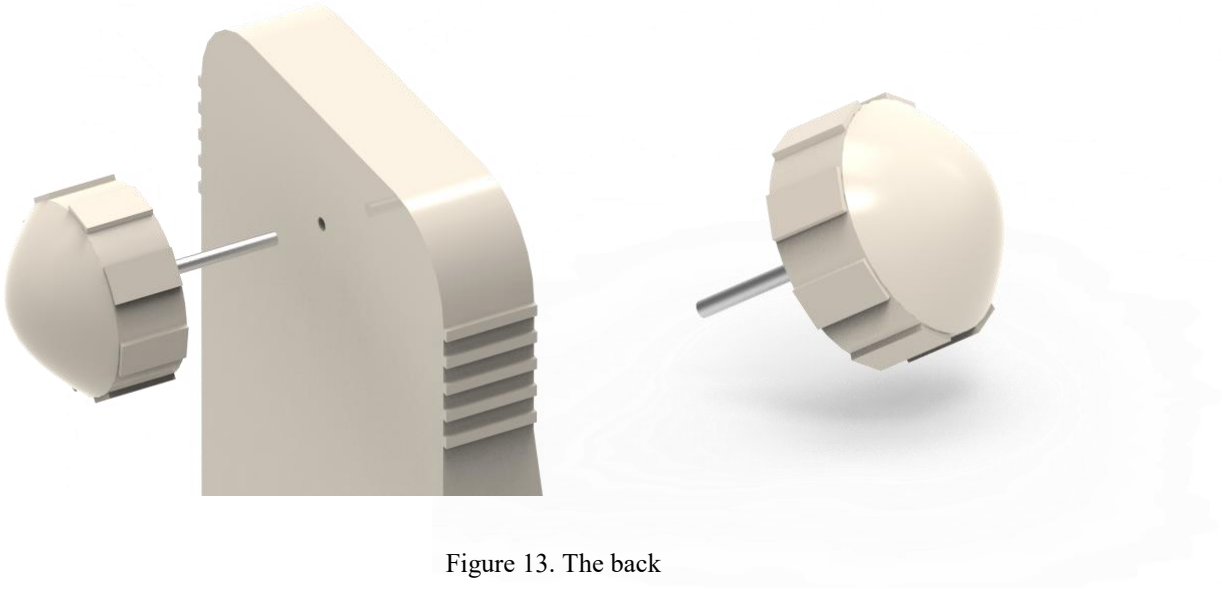


Figure 13. The back

As shown in Figure 13 on the right, the grooves run all the way around, which enhance grip regardless of where the product is held on the back. The rounded top is an important feature that increases the tactile comfort against the skin when customers try on tight-fitting garments.

Figure 14 shows the final product featuring a price and product information layout. The design shown serves as an example of how the interface could look like. However, the intention is for each store to have the flexibility to customize the layout to their liking, in order to meet their particular needs.



Figure 14. The product

7.2 Evaluation against requirements specification

All requirements and desired features from the specification were reviewed, and it was decided whether each criterion had been fulfilled or not. In Appendix I, these have been marked in green or red depending on whether they were achieved or not. The appendix also includes a column containing the letters “K” and “Ö”, where “K” stands for requirements and indicates criteria that must be fulfilled, while “Ö” stands for desired features, meaning they are not mandatory but considered beneficial if achieved.

As shown in Appendix I, all requirements (K) were fulfilled, along with the majority of the desired features (Ö). The group concluded that it was acceptable for some desired features not to be fully achieved, as the overall solution was still considered advanced, functional, and well-developed.

7.3 Manufacturing

The manufacturing process is based on injection moulding, which is used for both the front and back part. Each main component is moulded in two separate halves, meaning that the front part consists of two individual moulded pieces, and the same applies to the back part, resulting in a total of four parts. All electronics and technology are assembled after the injection moulding process, and finally, the two halves of both the front and back part are glued together to form the complete product. The manufacturing dimensions are included in Appendix J.

8 DISCUSSION

This chapter presents a discussion of key topics from the project.

The goal for this project was to develop a product that solves the problems identified in the analysis.

As a result, features enhancing the store operations were prioritized, even though they may increase the overall costs. In a real-world market scenario, the economic side of it would play a larger role. A product like this one has to be cost efficient to profit, and cannot only focus on the functionality of the product. For this project, it was decided to maximize the functions, but in some cases, that is not what the retailers are looking for.

One example of this is the NFC technology implemented to streamline checkout operations. Although NFC enables faster checkouts, it comes with higher costs. However, cost has been a main discussion point. The supervisor explained that in this case, since the product does not yet exist on the market, cost should not limit the outcome of the project. Instead, the focus should be on creating a strong solution first, based on the findings from the analysis phase. In a future development, the focus could be shifted trying to find the cheapest solution possible, while still implementing necessary technology, similar to Concept 2 (The Minimalist) described in Concept Development.

Earlier in the report, it was researched and concluded that all the electronics and components necessary would fit in the tag. However, in a continued development of this project, an investigation on how these specific components would be placed internally and wired would have to be made.

One topic for discussion is also readability. During the report, it has been described how an angle of the screen makes it easier for the customer to read the displayed information. However, this goes under the assumption that the tag is placed vertically on the garment, shown fully and not hidden under a collar or similar.

In addition, design for assembly DFA and design for recycling DFR would require further investigation. With several electronic components integrated, this has to be further developed.

While the solution reduces paper waste, it integrates electronic components which can introduce environmental concerns. Both opportunities and challenges are created and the trade-off has to be weighed in order to reach a solution which is sustainable.

Further sustainability analysis is therefore needed. A LCA would give a better understanding of the product's total environmental impact compared to existing retail security solutions. In addition, factors such as repairability, recyclability, and durability in demanding retail environments should be further explored. Since the product will be used daily and exposed to wear, impacts, scratches, and frequent handling, it is important that it has a long lifespan to justify the environmental impact of its production.

9 CONCLUSION

The conclusion answers the problem statement and gives recommendations for further development.

The project has resulted in a modern, technologically advanced product concept called the Smart Reusable Retail Security Tag, designed to compete with and improve upon today's retail security solutions. A solution that integrates theft protection, price management, and cash register systems, which streamlines store operations while maintaining a premium experience for the customers.

The development process showed that advanced technologies such as NFC, RFID and e-paper displays can create a more efficient retail experience. At the same time, the project showed the importance of weighing economics in order to reach a product that would attract stores.

Although the concept remains at a conceptual stage, the project identified several important areas for future development, including internal component integration, construction, and long-term durability testing. Furthermore, more sustainability evaluations, such as LCA and recyclability studies, are needed to fully understand the environmental impact of the solution.

Overall, the project shows that a reusable smart retail tag could optimize retail stores by streamlining the work performed by retail employees, leading to reduced labor costs, improved inventory data, and faster checkout processes. This could represent a milestone in the industry, potentially redefining how retail stores operate by making them more modern, efficient, and technology driven.

10 SOURCES

Atlas RFID (n.d). What is RFID? How RFID system work. https://www.atlasrfidstore.com/rfid-resources/rfid-beginners-guide/?srsltid=AfmBOorX_s-CHlS-jbWnqUrUQyJMbh4M2PgUnm1iHUgXTm1RXHwsshGk (Accessed 2026-04-07)

Dagens Logistik (n.d). Handelns starkast lysande stjärnor i nio olika priskategorier hyllades när galan Retail Awards igår gick av stapeln för femtonde gången. Stadium kammade hem priset ”Årets logistiklösning” för sitt införande av RFID-teknik. https://dagenslogistik.se/stadium-kammade-hem-handelns-logistikpris/ (Accessed 2026-04-08)

Fortune Business Insights (2026). RFID Market Size, Share & Industry Analysis, By Product Type (Tags, Readers, and Software & Services), By Frequency (Low Frequency, High Frequency, and Ultra-High Frequency), By Industry (Agriculture & Forestry, Healthcare, Manufacturing, Retail, Transportation and Logistics, and Others), and Regional Forecast, 2026–2034. https://www.fortunebusinessinsights.com/rfid-market-109243? (Accessed 2026-04-07)

Huayuan (n.d). RFID Case Study: UNIQLO Global Stores Applied RFID Tags. https://huayuansh.com/uniqlo-global-stores-applied-rfid-tags/ (Accessed 2026-04-07)

Lindholm, A (2026). Hållbarhet i varumärkesbyggande: Mer än bara en hygienfaktor? https://www.ipsos.com/sv-se/hallbarhet-i-varumarkesbyggande-mer-bara-en-hygienfaktor (Accessed 2026-04-08)

Pricer (n.d). Optic Wireless Network. https://www.pricer.com/products/technology/optical-wireless-network (Accessed 2026-04-07)

Wikipedia (2025). Närfältskommunikation. https://sv.wikipedia.org/wiki/N%C3%A4rf%C3%A4ltskommunikation (Accessed 2026-04-07)

11.1 Appendix A: Store visits





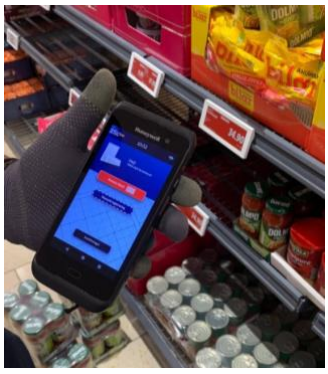












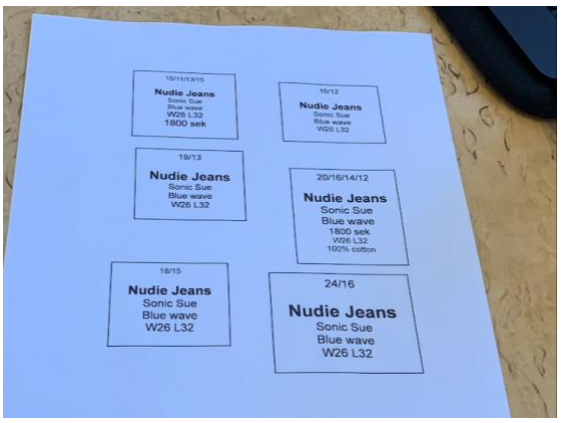
11.2 Appendix B: Requirement specifications

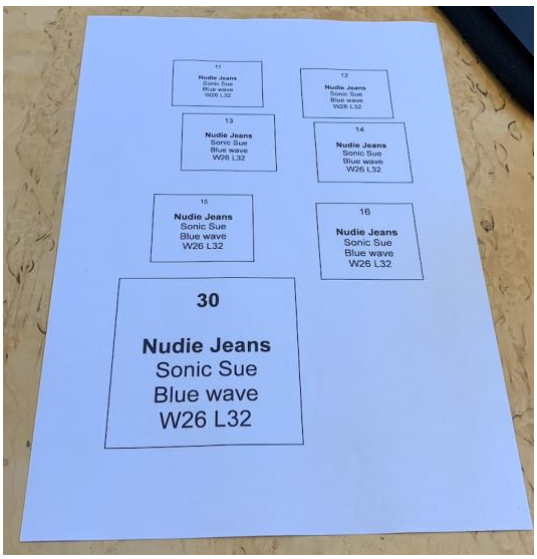
Function	
Can be updated daily	K
Can be attached to a garment	K
Integrated anti-theft protection	K
Clear e-paper display	K
Simplifies inventory management	Ö
More efficient checkout process	Ö
Increased inventory accuracy	Ö
Technical requirements	
Integrated RFID or NFC	K
Ability to update RFID/NFC information	K
Battery life of at least 5 years	K
Thin battery for minimal bulk	Ö
Resistant to basic conditions such as moisture and dust	K
Ability to see battery status	Ö
Magnetic locking mechanism (EAS)	K
Material	
Provides a premium feel	Ö
Recyclable	K
Pleasant tactile feel against the skin	Ö
Other	
Perceived as modern	Ö
DFA & DFR	Ö
Semiotics for easier usability	Ö

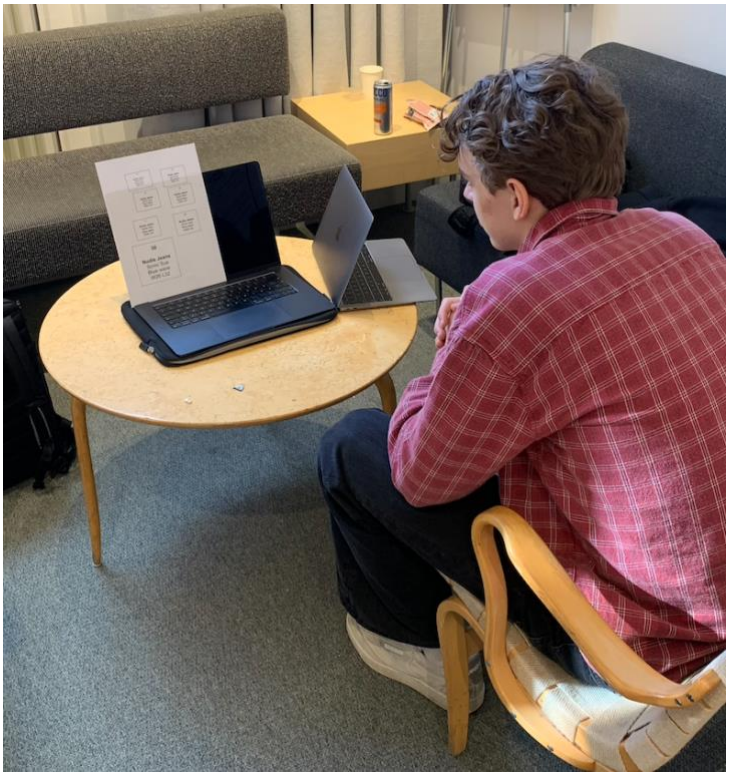
11.3 Appendix C: Revisits to stores



11.4 Appendix D: Visibility test







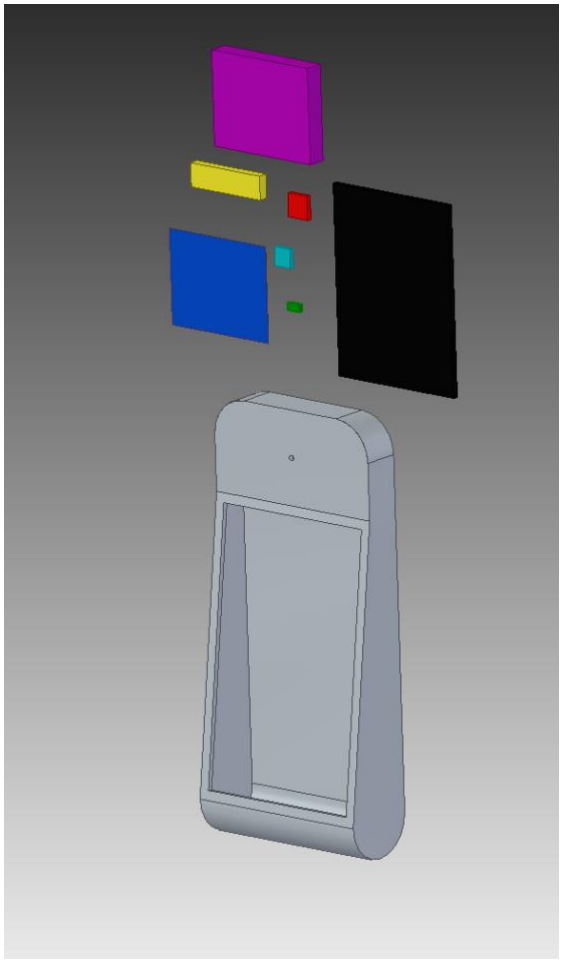
11.5 Appendix E: Validation of the screen angle

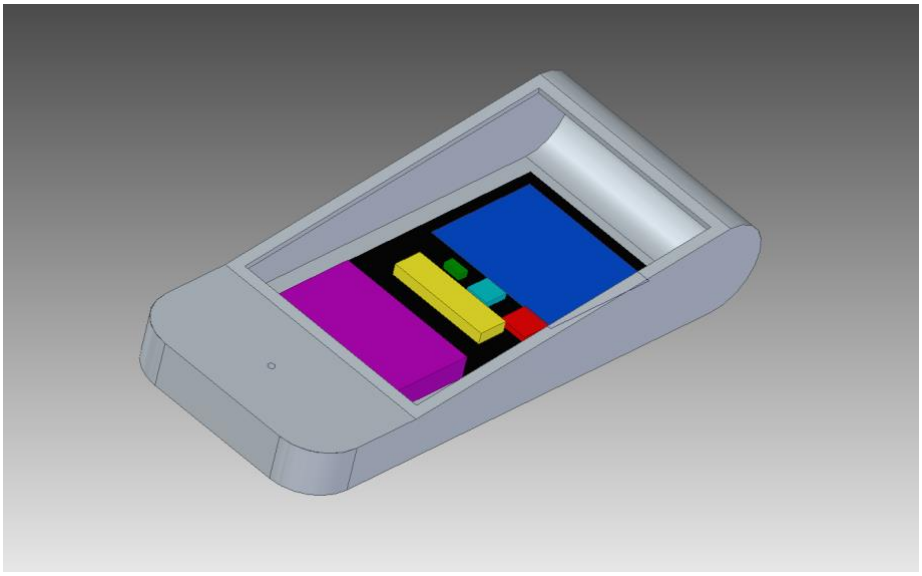


11.6 Appendix F: Testing the back component



11.7 Appendix G: Analysis of internal layout





11.8 Appendix H: Cost and weight calculation

Components	**Price**	**Weight**
Microcontroller	15 kr	1,5 g
2,4 GHz antenna	5 kr	0,5 g
Printed circuit board	5 kr	6 g
NFC antenna	5 kr	1 g
RFID chip	1 kr	0,1 g
Passive components	3 kr	0,5 g
E-paper display	50 kr	15 g
FPC connector	1,5 kr	0,5 g
Battery	4 kr	3 g
Housing	1 kr	30 g
EAS anti-theft system	5 kr	5 g

11.9 Appendix I: Requirements specification validation

Function		
Can be updated daily	K	
Can be attached to a garment	K	
Integrated anti-theft protection	K	
Clear e-paper display	K	
Simplifies inventory management	Ö	
More efficient checkout process	Ö	
Increased inventory accuracy	Ö	
Technical requirements		
Integrated RFID or NFC	K	
Ability to update RFID/NFC information	K	
Battery life of at least 5 years	K	
Thin battery for minimal bulk	Ö	
Resistant to basic conditions such as moisture and dust	K	
Ability to see battery status	Ö	
Magnetic locking mechanism (EAS)	K	
Material		
Provides a premium feel	Ö	
Recyclable	K	
Pleasant tactile feel against the skin	Ö	
Other		
Perceived as modern	Ö	
DFA & DFR	Ö	
Semiotics for easier usability	Ö	

11.10 Appendix J: Manufacturing

